

Beginning Algebra Practice Test I Version 1

Name
Date
Course

Instructions

- i) As always, write a title, name, date, and course.
- ii) Number each problem, circle the numbers, and show work when necessary.
- iii) Draw a rectangle around your answers.
- iv) Staple a copy of these test questions with your work and your final answers.

Simplify the following polynomial, write in descending order of powers, and state its degree.

① $-x^4 + 7 - 6x^2 + 2 - 4x^2 + 3x^4 - 7x - 8 + x^3$
 $2x^4 + x^3 - 10x^2 - 7x + 1$, $D=4$

② Add the following polynomials.

a) $(-5x^3 + 3x^2 - x + 4) + (8x^3 + 9x - 1)$
 $-5x^3 + 3x^2 - x + 4 + 8x^3 + 9x - 1$
 $3x^3 + 3x^2 + 8x + 3$

b) $(2x - 7) + (x^2 - 3x - 3)$

$$2x - 7 + x^2 - 3x - 3$$

$$x^2 - x - 10$$

③ Subtract the following polynomials.

a) $(4x^2 + 2x - 6) - (3x^2 - 7x - 12)$

$$4x^2 + 2x - 6 + (-3x^2) + 7x + 12$$

$$x^2 + 9x + 6$$

b) $(10x^3 + 4x - 7) - (-x + 2)$

$$10x^3 + 4x - 7 + x - 2$$

$$10x^3 + 5x - 9$$

Multiply.

④ a) $-4x(3x^2 + 9x - 4)$ b) $6x^3(-3x^3 + 4x^2 - x + 1)$

$$-12x^3 - 36x^2 + 16x$$

$$-18x^6 + 24x^5 - 6x^4 + 6x^3$$

Divide.

⑤ a) $\frac{-32x^4 - 16x^3 + 56x^2}{-8x^2}$

$$4x^2 + 2x - 7$$

b) $\frac{63x^4 + 35x^3 - 56x^2}{7x}$

$$9x^3 + 5x^2 - 8x$$

Expand.

⑥ a) $(x-4)(2x-1)$ b) $(3x+5)(5x-6)$
 $2x^2 - x - 8x + 4$ $15x^2 - 18x + 25x - 30$
 $\boxed{2x^2 - 9x + 4}$ $\boxed{15x^2 + 7x - 30}$

⑦ a) $(7a+2b)(3a+b)$ b) $(x-2)(6x^2+3x-1)$
 $21a^2 + 7ab + 6ab + 2b^2$ $6x^3 + 3x^2 - x - 12x^2 - 6x + 2$
 $\boxed{21a^2 + 13ab + 2b^2}$ $\boxed{6x^3 - 9x^2 - 7x + 2}$

⑧ a) $(x+7)^2$ **FOIL** b) $(2x-y)^2$
 $(x+7)(x+7)$ $(2x-y)(2x-y)$
 $x^2 + 7x + 7x + 49$ $4x^2 - 2xy - 2yx + y^2$
 $\boxed{x^2 + 14x + 49}$ $\boxed{4x^2 - 4xy + y^2}$

Factor the following expressions completely.

⑨ a) $x^2 - 2x - 15$ b) $4x^2 + 8x - 21$
 $\boxed{(x+3)(x-5)}$ $\boxed{(2x-3)(2x+7)}$

⑩ a) $a^2 - 8ab + 12b^2$ b) $x^2 - 4$
 $\boxed{(a-6b)(a-2b)}$ $\boxed{(x+2)(x-2)}$

⑪ a) $9x^2 - y^2$ b) $-36 + x^2$
 $\boxed{(3x+y)(3x-y)}$ $x^2 - 36$
 $\boxed{(x-6)(x+6)}$

⑫ a) $x^2 - 100$ b) $16 - y^2$
 $\boxed{(x-10)(x+10)}$ $\boxed{(4+y)(4-y)}$

⑬ a) $-x^2 + 81$ b) $49 - 64b^2$
 $81 - x^2$ $\boxed{(7+8b)(7-8b)}$
 $\boxed{(9-x)(9+x)}$

⑭ a) $x^4 + 3x^2 - 4$ b) $x^4 - 20x^2 + 64$
 $(x^2+4)(x^2-1)$ $\boxed{(x^2-4)(x^2-16)}$
 $\boxed{(x^2+4)(x-1)(x+1)}$ $\boxed{(x-2)(x+2)(x-4)(x+4)}$

$$\textcircled{15} \text{ a) } 6x^4 + 7x^2 - 3$$

$$(2x^2 + 3)(3x^2 - 1)$$

$$\text{b) } 10x^3 - 80x^2 + 3x - 24$$

$$10x^2(x - 8) + 3(x - 8)$$

$$(x - 8)(10x^2 + 3)$$

$$\textcircled{16} \text{ a) } x^3 - 3x^2 - x + 3$$

$$x^2(x - 3) - 1(x - 3)$$

$$(x - 3)(x^2 - 1)$$

$$(x - 3)(x - 1)(x + 1)$$

$$\text{b) } b^5 + 2b^4 - 8b - 16$$

$$b^4(b + 2) - 8(b + 2)$$

$$(b + 2)(b^4 - 8)$$

$$\textcircled{17} 7x^3 - 3x^2 - 14x + 6$$

$$x^2(7x - 3) - 2(7x - 3)$$

$$(7x - 3)(x^2 - 2)$$

$$\textcircled{18} 3x^3 - 75x$$

$$3x(x^2 - 25)$$

$$3x(x - 5)(x + 5)$$

$$\textcircled{19} 9x^3 + 18x^2 + 9x$$

$$9x(x^2 + 2x + 1)$$

$$9x(x + 1)(x + 1)$$

$$\textcircled{20} 24x^8 - 54x^4$$

$$6x^4(4x^4 - 9)$$

$$6x^4(2x^2 - 3)(2x^2 + 3)$$

$$\textcircled{21} -3x^4 + 312x^2 - 1200$$

$$-3(x^4 - 104x^2 + 400)$$

$$-3(x^2 - 4)(x^2 - 100)$$

$$-3(x - 2)(x + 2)(x - 10)(x + 10)$$

$$\textcircled{22} 32a^3b - 18ab^3$$

$$2ab(16a^2 - 9b^2)$$

$$2ab(4a - 3b)(4a + 3b)$$

$$\textcircled{23} -45x^3 + 42x^2y - 9xy^2$$

$$-3x(15x^2 - 14xy + 3y^2)$$

$$-3x(3x - y)(5x - 3y)$$

$$\textcircled{24} -2x^3 + 3x^2 + 2x - 3$$

$$-x^2(2x-3) + 1(2x-3)$$

$$(2x-3)(-x^2+1)$$

$$(2x-3)(1-x^2)$$

$$(2x-3)(1-x)(1+x)$$

$$\textcircled{25} 12a^4b^3 - 2a^3b^4 - 10a^2b^5$$

$$2a^2b^3(6a^2 - ab - 5b^2)$$

$$2a^2b^3(6a+5b)(a-b)$$